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GitHub repository: https://github.com/Xuanxuan-zi/ABM_Rebecca_final

Cultural Boundaries, Language Capital, and Institutional Contact: An Agent Based Model of International Student Segregation

Abstract

This paper develops an original agent based model to study how cultural distance, English language barriers, stress, and university supported contact shape social segregation among international students on a multicultural campus. Instead of treating international student clustering as a simple individual preference, the model examines how segregation can emerge from repeated local interactions, unequal language resources, stress accumulation, and different access to comfortable cross cultural contact. Each student agent has a country label used for initialization, a five dimensional cultural vector, English fluency, stress level, conflict memory, and maintained social ties. I compare three population scenarios with and without a structured pairing intervention similar to a buddy program. Batch runs show that the intervention increases cross cultural friendship share and reduces network homophily, but it does not eliminate spatial clustering. The model suggests that university programs may reshape friendship networks more easily than everyday spatial comfort zones.

1. Introduction

Universities often describe international diversity as evidence of openness and global connection. A campus may include students from many countries, and these students may share classrooms, libraries, and dining spaces. However, physical co presence does not automatically create social integration. In everyday campus life, students may still rely on familiar language groups, spend time with culturally similar peers, and form friendship networks divided by national or cultural background (Moody, 2001).

This paper studies the gap between demographic diversity and social integration. The main research question is: Can a university supported structured pairing intervention reduce cultural segregation among international students, and does its effect appear more strongly in friendship networks than in spatial clustering? A second question is how population composition shapes the opportunity structure for cross cultural friendship.

The model is motivated by a sociological view of segregation. Segregation is not always produced by open hostility or explicit exclusion. It can also emerge through small repeated decisions under unequal conditions. A student with limited English fluency may find cross country interaction more tiring (Smith et al., 2011). A student who has experienced stressful interactions may become more likely to stay near familiar peers (Currarini et al., 2009). A large same country group may provide more opportunities for comfortable interaction inside the group. Over time, these micro level processes can produce macro level clustering and network homophily.

The model also asks what universities can do. Buddy programs, conversation partner programs, and intercultural orientation groups are based on the idea that contact can support integration.

However, random proximity may not be enough if students are placed together without support. For this reason, the model separates ordinary spatial contact from structured institutional contact. The main argument is that university intervention can weaken network level segregation, even if it does not fully undo spatial clustering.

2. Literature Review and Framing

Research on segregation and homophily helps explain why students may remain socially divided even in formally diverse settings. Schelling (1971) shows that aggregate segregation can emerge from mild local preferences rather than explicit exclusion. Network research also shows that friendship formation is strongly shaped by similarity and opportunity structure (McPherson, Smith-Lovin, and Cook, 2001; Currarini, Jackson, and Pin, 2009). In school settings, Moody (2001) shows that formal integration does not necessarily remove friendship segregation. These studies are useful for this project because they show that segregation does not have to come from open prejudice. It can also develop from repeated small choices about where students spend time, whom they interact with, and which relationships are easier to maintain.

This logic is especially relevant for international students because cross-cultural interaction may involve different costs for different students. Acculturation research shows that adaptation involves both keeping one's cultural background and having contact with other groups (Berry, 1997). Ward and Kennedy (1999) distinguish psychological adjustment from sociocultural adjustment, which is useful for this model because students may be physically present on campus but still struggle with everyday interaction. Research on international students also shows that English fluency, social connectedness, and stress are important for student adjustment (Yeh and Inose, 2003; Smith and Khawaja, 2011). These studies support the use of language and stress in the model. In the model, English fluency affects whether cross-cultural interaction is easy or difficult. Stress then affects whether students continue to interact, move toward familiar spaces, or become isolated.

Contact theory adds another important point. Contact is more likely to be effective when it happens under supportive conditions. Allport (1954) argues that intergroup contact works better when there is cooperation, shared goals, and institutional support. Pettigrew and Tropp (2006) also show that contact can reduce prejudice, but this does not mean that random exposure is enough. Evidence from college settings also suggests that exposure to diversity can shape later attitudes and social outcomes, but these effects should not be treated as automatic consequences of physical co-presence (Boisjoly et al., 2006). This matters for the model because students may share the same campus spaces without forming meaningful cross-cultural friendships. A classroom, dining hall, or library can create ordinary contact, but that contact may still feel awkward or tiring. For this reason, the model separates ordinary spatial contact from structured university-supported contact.

This paper brings these ideas together in an agent-based model. The model does not assume that students simply prefer people from their own country. Instead, it shows how segregation can grow from repeated interaction costs. Cultural distance, English fluency, stress, movement, and tie maintenance are connected in the model. Difficult interaction can increase stress. Stress makes familiar spaces more attractive. Repeated comfortable interaction can create stronger social ties. Ties that are not reinforced gradually decay. Through these repeated steps, individual-level interaction experiences can produce spatial clustering and network homophily.

The model also separates two outcomes: spatial mixing and friendship-network integration. This is important because a university program may change who becomes friends with whom without fully changing where students spend everyday time. The R5 intervention is used to model structured university-supported contact. It tests whether supported cross-cultural pairing can reduce network homophily more clearly than spatial clustering.

Cultural distance is used only as a modeling tool. Hofstede-style dimensions allow the model to represent cultural difference as a continuous distance rather than treating all cross-national contact as the same (Hofstede, 2001). Similar work using cultural distance measures also shows the usefulness of representing cross-national differences by degree rather than as simple categories (Shulgin et al., 2017). At the same time, the model follows McSweeney's (2002) warning that national culture should not be treated as fixed or homogeneous. Country labels are used only to initialize cultural vectors and language ranges. Individual agents still vary within each country group, and their cultural vectors can change slightly through interaction.

3. Social Mechanism

The central mechanism of the model is that small interaction costs can gradually become social structure. The model does not directly assign segregation to agents. Instead, segregation emerges through repeated local encounters, stress responses, movement choices, and tie maintenance.

Students first meet nearby others. If an interaction is culturally familiar and linguistically easy, it reduces stress and can help create or strengthen a maintained social tie. If the interaction involves moderate cultural distance, students may learn from each other, but the interaction also carries a small stress cost. If cultural distance is high, especially when effective fluency is low, the interaction increases stress and conflict memory.

Stress then affects later behavior. Active students choose among nearby empty cells, but spaces with more culturally similar neighbors are more attractive. When stress is high, this preference for familiar surroundings becomes stronger. As a result, small local clusters can become self-reinforcing over time. Social ties follow a similar logic. Repeated successful encounters strengthen ties, while ties that are not reinforced gradually decay. This means that integration requires repeated relationship maintenance, not only one-time contact.

The R5 intervention interrupts this process by creating structured cross-country contact. It reduces stress, supports small cultural adaptation, improves English fluency, and strengthens cross-cultural ties. Because R5 directly supports tie formation but only mildly changes movement, I expect it to reduce network homophily more clearly than spatial clustering.

4. Model Design and Operationalization

This section translates the social mechanism into operational model rules. The model first defines the population composition, agent attributes, grid environment, and tick schedule. It then specifies how R1–R3 update stress, learning, conflict memory, and social ties; how R4 determines movement; and how R5 implements structured university-supported contact. The detailed rule conditions and parameter values are reported in Tables 1–5.

4.1 Population Scenarios

The model includes three population scenarios. These scenarios are used to vary the demographic opportunity structure for cross-cultural friendship. They are not meant to represent all possible international student populations.

Table 1. Population Scenarios

Scenario	U.S.	China	India	S. Korea	Germany	Brazil
Baseline	50%	15%	12%	8%	8%	7%
Chinese heavy	30%	50%	5%	5%	5%	5%
Balanced	25%	25%	25%	10%	10%	5%

4.2 Agent Initialization and Tick Schedule

After the population composition is defined, agents are initialized with country labels, cultural vectors, English fluency, stress levels, and social-tie records. Table 2 also reports the tick schedule, which shows the order in which tie decay, movement, interaction, stress updating, intervention, and data collection occur.

Table 2. Model Setup, Agent Initialization, and Tick Schedule

Component	Operationalization	Parameter / value
Agent type	Each agent represents one student.	—
Number of agents	Main model runs use a fixed population size.	500
Space	Agents occupy a 50 × 50 toroidal grid. Each cell can contain at most one agent.	50 × 50; torus = true
Country assignment	Each agent receives a country label according to the selected population scenario. Country counts are drawn from a multinomial distribution.	Baseline / Chinese-heavy / Balanced

Initial location	Agents are randomly placed on empty grid cells.	Random empty cells
Cultural vector	Each agent receives a five-dimensional cultural vector initialized around a country-specific center. Random normal noise is added and values are clipped to [0, 1].	$\sigma = 0.08$; range = [0, 1]
Cultural distance	Distance is the normalized Euclidean distance between two agents' cultural vectors.	Similar < 0.30; moderate = 0.30–0.45; high > 0.45
English fluency	Initial English fluency is drawn from country-specific ranges.	U.S. 0.95–1.00; China 0.40–0.80; India 0.70–0.95; S. Korea 0.50–0.80; Germany 0.75–0.95; Brazil 0.50–0.80
Effective fluency	Same-country pairs are assumed to share enough linguistic background for ordinary interaction. Cross-country pairs use the lower English fluency of the two students.	Same-country = 1.00; cross-country = min(fluency A, fluency B)
Initial individual state	All agents start active with no stress or conflict memory.	Stress = 0.00; conflict memory = 0.00; state = active; recovery counter = 0
Initial social state	Encounter histories and structured-contact partner sets are empty. The friendship network starts with all agents as nodes and no ties.	No initial social ties
Tick order	At each tick, existing ties decay first. Agents then act in random order. Active agents move, interact with active neighbors, receive passive language stress if applicable, and update stress recovery. R5 is applied after ordinary movement and interaction. Data are collected at the end of the tick.	Fixed schedule
Double-counting rule	Each dyadic pair can interact at most once per tick.	One interaction per pair per tick

4.3 Interaction, Stress, and Tie Maintenance

The main interaction process is organized through R1–R3. These rules determine how cultural distance and effective fluency change stress, learning, conflict memory, and social ties during local dyadic encounters.

Table 3. Interaction Rules R1–R3, Stress Updates, and Tie Maintenance

Rule	Condition	Agent-level update	Key parameter values
R1 Similar interaction	Cultural distance < 0.30 and effective fluency ≥ 0.30	Stress decreases for both agents. Encounter count increases by 1. Repeated encounters can create or strengthen a social tie.	Stress reduction = 0.05 × effective fluency; base friendship threshold = 12; required encounters = ceiling(12 / max(effective fluency, 0.30)); tie increment = 0.35
R2 Learning interaction	0.30 ≤ cultural distance ≤ 0.45	Learning occurs only if the interaction succeeds. If successful, cultural vectors move slightly closer. For cross-country pairs, English fluency increases. Stress also rises slightly.	Success probability = effective fluency; learning rate = 0.002; language gain = 0.005; stress cost = 0.01
R3 Conflict interaction	Cultural distance > 0.45	Stress and conflict memory increase for both agents. If stress reaches the isolation threshold, the agent becomes isolated.	Base stress = 0.10; stress increase = 0.10 × distance factor × memory multiplier × language penalty; memory increment = 0.05; isolation threshold = 0.85
Passive language stress	Agent fluency < 0.70 and nearby foreign/culturally different neighbors are present	Background stress increases due to a linguistically demanding local environment.	Passive stress = 0.03 × foreign-neighbor ratio × fluency deficit
Stress lifecycle	Every tick	Stress and conflict memory decay gradually. Isolated agents can return to active status after several low-stress ticks.	Stress decay = 0.98; conflict memory decay = 0.995; recovery threshold = 0.40; recovery duration = 5 ticks

Tie lifecycle	Every tick	Unreinforced social ties lose strength. Very weak ties are removed.	Tie decay = 0.985; removal threshold = 0.25; maximum tie strength = 4.00
Rule assignment	Every local dyadic interaction	R1, R2, and R3 are mutually exclusive. Each interaction is assigned to one rule based on cultural distance.	One rule per dyadic interaction

4.4 Movement Rule

Rule R4 connects interaction experience to spatial behavior. It specifies how active agents choose nearby empty cells and how stress increases the attractiveness of culturally familiar spaces.

Table 4. Movement Rule R4

Agent condition	Movement rule	Operational detail
Isolated agent	Does not move or interact.	The agent only updates stress recovery.
Active agent with no empty neighboring cell	Stays in place.	No movement occurs if all nearby cells are occupied.
Active agent under normal stress	Considers all empty Moore-neighborhood cells.	Candidate cells with more culturally similar neighbors receive higher movement weight.
Active agent under high stress	Uses the same candidate cells, but the preference for culturally similar neighbors becomes stronger.	High stress is defined as stress > 0.50.
Active agent near R5 partner	Candidate cells near structured-contact partners receive a small movement bonus.	This creates a weak spatial pull toward R5 partners but does not erase ordinary clustering.
Final movement choice	Agent randomly selects one candidate cell using weighted probabilities.	Probability of choosing a cell equals its weight divided by the sum of all candidate-cell weights.

Weight formula	A culturally similar neighbor is defined as a neighbor with cultural distance below 0.30.	Baseline weight = 1.3^pairs, where pairs = floor(similar_neighbors / 2). If stress > 0.50, the same multiplier is applied again. If structured-contact partners are nearby, the cell receives 1.15^structured_neighbors.

4.5 Structured Pairing Intervention

Rule R5 models the structured university intervention. Unlike ordinary spatial contact, R5 creates supported cross-country pairings under manageable conditions and directly strengthens cross-cultural ties.

Table 5. R5 Structured Pairing Intervention

Component	Operation	Parameter value
Intervention condition	R5 runs only when the intervention condition is enabled.	On/off
Timing	Runs every 20 ticks, excluding tick 0.	Frequency = 20
Eligible agents	Only active agents can be selected.	Isolated agents excluded
Coverage	Each cycle reaches about 20% of the student population.	Coverage rate = 0.20
Number of pairs	In the 500-agent model, each cycle can create up to 50 pairs.	Pairs per event = 50
Pair uniqueness	Each student can appear at most once in the same intervention cycle.	Unique student constraint
Pair type	Pairs must be cross-country.	Same-country pairs rejected
Cultural distance rule	Pairs must be different enough for cross-cultural learning but not too distant to be unrealistic.	$0.30 < \text{distance} \leq 0.55$

Fluency rule	The pair must have enough effective fluency for supported contact.	Effective fluency ≥ 0.40
Encounter effect	A structured program cycle counts as several meaningful encounters.	Encounter boost = 4
Stress effect	Stress decreases for both paired agents.	Stress reduction = 0.05
Cultural adaptation	The two cultural vectors move slightly toward each other.	Learning rate = 0.006
Language effect	English fluency increases for both agents, capped at 1.00.	Language gain = 0.006
Tie effect	If the boosted encounter count reaches the R5 threshold, a social tie is created or strengthened.	Friendship threshold = 4; tie increment = 0.90
Movement side effect	The partner is stored as a structured-contact partner.	Future movement bonus = 1.15 per nearby structured partner
Search limit	The model attempts to find eligible pairs but stops after a maximum number of attempts.	Max attempts = target pairs × 100
Interpretation	R5 represents structured university-supported contact. It is intentionally stronger than ordinary contact for tie formation, but its spatial effect is mild.	Network effect > spatial effect

Together, these operational rules make the model reproducible. They show how individual-level differences in cultural distance, English fluency, stress, and repeated contact can produce aggregate outcomes such as spatial clustering and network homophily.

5. Experimental Design

The experiment explores the model's dynamics by varying two main parameters: population composition and intervention status. Population composition takes three values: baseline, Chinese-heavy, and balanced. Intervention status takes two values: no intervention and R5 structured pairing. This produces six model conditions in total. Other parameters, including grid size, number of agents, interaction thresholds, stress update rates, movement weights, and tie decay rates, are held fixed at the values reported in Tables 1–5. This design isolates the effects of demographic opportunity structure and structured university-supported contact. Each condition is repeated 20 times with different random seeds, and each run stops at tick 300.

The main outcomes are spatial clustering, network homophily, cross cultural friendship share, mean stress, isolation rate, active social ties, mean tie strength, and mean cultural distance. Spatial clustering measures the share of culturally similar neighbors around agents. Network homophily measures the share of active social ties connecting students from the same country. Cross cultural friendship share is one minus network homophily. Mean stress and isolation rate measure the welfare side of the model. Active social ties measure the size of the maintained friendship network.

I expect the intervention to increase cross cultural friendship share and reduce network homophily. I also expect stress and isolation to decline. I do not expect spatial clustering to disappear, because the intervention directly targets structured contact and social ties rather than completely changing movement preferences.

6. Results

The batch results show that the intervention affects friendship networks more clearly than spatial clustering. Each scenario and intervention condition was repeated 20 times and compared at tick 300. Structured pairing increases cross cultural friendship share and reduces network homophily in all three population scenarios, as shown in Figure 1.

The increase in cross cultural friendship share is largest in the Chinese heavy scenario. It rises from 36.5 percent without intervention to 47.7 percent with intervention, an increase of 11.2 percentage points. The baseline scenario also increases from 51.4 percent to 57.7 percent, while the balanced scenario increases from 67.1 percent to 71.7 percent. This pattern is sociologically meaningful because the Chinese heavy scenario gives students more demographic opportunity to form same group ties. When one group is numerically dominant, structured university contact has more room to change the composition of the network.

Network homophily moves in the opposite direction. It falls by 6.3 percentage points in the baseline scenario, 11.2 percentage points in the Chinese heavy scenario, and 4.6 percentage points in the balanced scenario. This supports the main argument of the model. The intervention changes which kinds of ties remain active and makes cross cultural relationships more maintainable.

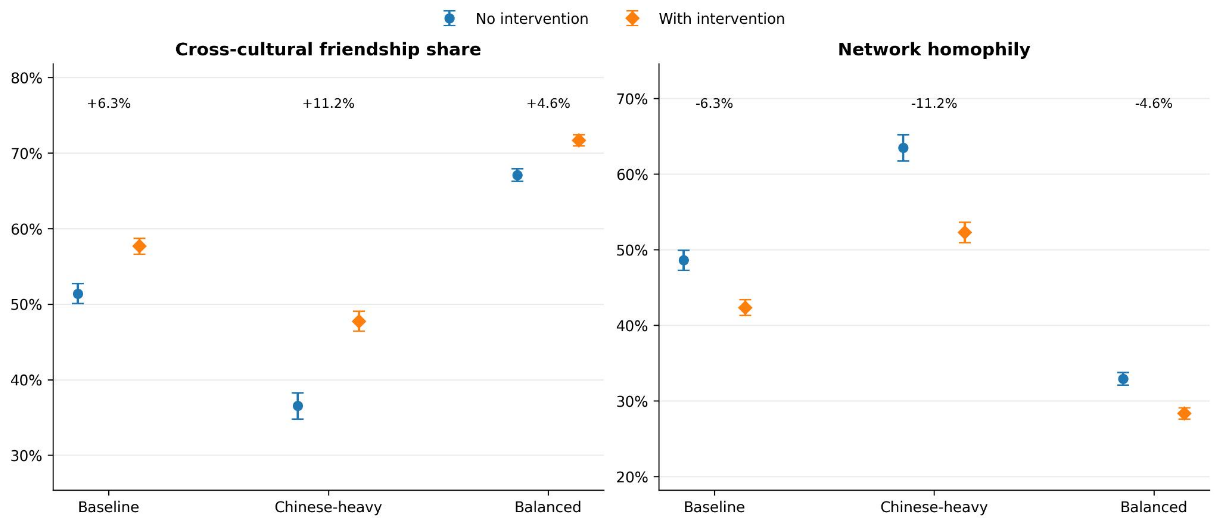

Figure 1. Network-Level Outcomes by Population Scenario and Intervention Condition

The direction and size of the intervention effects are summarized in Figure 2. In addition to changing friendship composition, the intervention increases the total number of active social ties. Active ties increase by about 258 in the baseline scenario, 229 in the Chinese heavy scenario, and 274 in the balanced scenario. Since ties in the model decay when they are not reinforced, this suggests that the intervention helps more relationships remain active by tick 300.

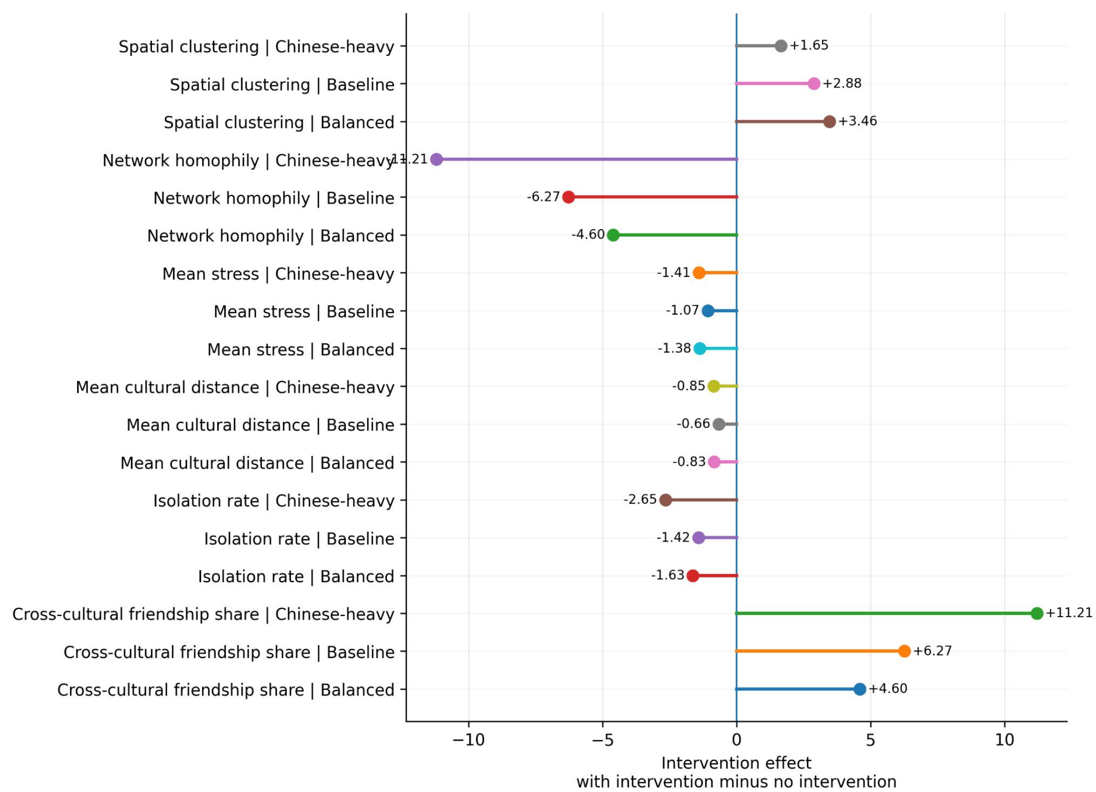

Figure 2. The Uneven Effects of University Intervention

The spatial results are more mixed. Spatial clustering does not decline after the intervention, and it slightly increases in all three scenarios, as shown in Figure 3. This does not necessarily mean that the intervention fails. Rather, it suggests that network integration and spatial clustering are different outcomes. The intervention directly supports cross cultural tie formation, but everyday movement is still shaped by cultural comfort, stress, and local surroundings.

The welfare outcomes also move in the expected direction. Mean stress decreases in all three scenarios, and isolation rate also declines, as shown in Figure 3. These changes are smaller than the network effects, but they are consistent with the model mechanism. Supported contact reduces some of the stress cost of cross cultural interaction and helps more students remain active in campus life.

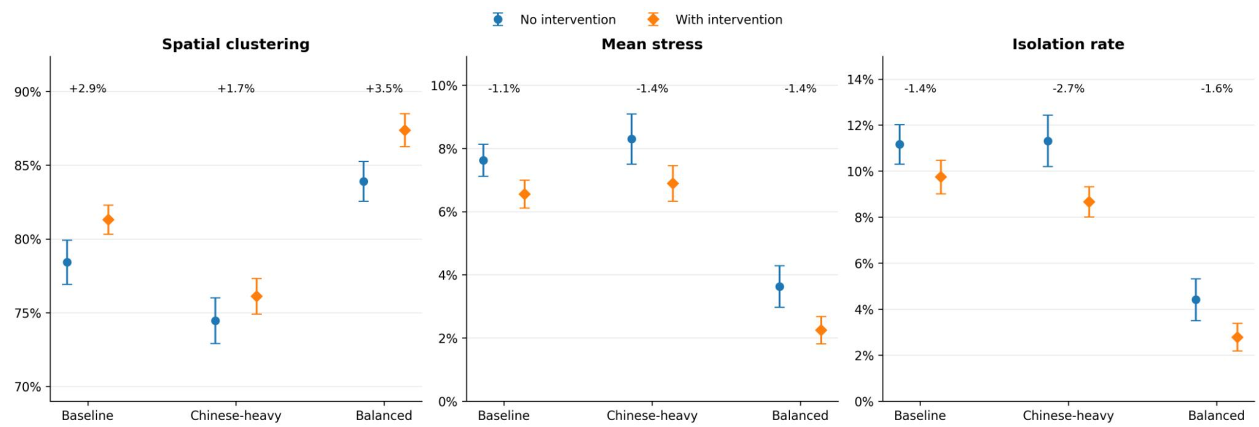

Figure 3. Spatial Clustering, Stress, and Isolation by Scenario

Overall, the results suggest that university supported contact can weaken network level segregation, but it does not fully eliminate spatial comfort zones. The main effect of the intervention is relational rather than spatial.

7. Discussion and Conclusion

These results suggest that campus integration should be understood as a layered process. The intervention changes friendship networks more clearly than spatial routines, which means that network integration and spatial integration should not be treated as the same outcome. A successful university program may create more cross cultural ties even when students continue to spend everyday time in familiar spaces.

This finding has policy implications. Universities should not assume that random proximity is enough. Structured programs matter because they lower the interaction cost of cross cultural contact and create repeated opportunities for tie maintenance. At the same time, the model suggests that these programs should be evaluated realistically. A buddy program may not erase all clustering. Its first effect may be to create bridges in friendship networks.

The model has several limitations. It is a stylized explanatory model, not an empirically calibrated simulation. Country labels and Hofstede style centers simplify real cultural variation. The campus grid also simplifies real campus spaces, such as dormitories, classrooms, clubs, dining halls, and online networks. The intervention is modeled as repeated structured pairing, but real programs vary in quality, duration, and student motivation. Future versions could compare stable long term buddy pairs with rotating group based programs.

Overall, the model helps reframe international student segregation as an emergent process shaped by language resources, stress, cultural boundaries, and opportunity structures. Institutional support can reshape these processes, but it does so unevenly. It may open friendship networks before it changes everyday spatial comfort zones.

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